Università della Svizzera italiana

Faculty of Informatics

Bioinformatics

Homework #3

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1 PSI-BLAST

Using the BLAST website, we can answer all the questions. Starting from the number of iteration, we can clearly see that we found the yeast sequence at the third iteration, and we found two of them:

- RecName: Full=RNA polymerase II subunit A C-terminal domain phosphatase; AltName: Full=CTD phosphatase FCP1 [Saccharomyces cerevisiae S288C]
- RecName: Full=Regulator of Ty1 transposition protein 107; AltName: Full=Establishes silent chromatin protein 4 [Saccharomyces cerevisiae S288C]

But we select the first one because have a e-value that is lower (3e-06), so basically is "more" homologous (true homology) compared to the second one (0.004).
We put here the screenshot of the sequence alignment, where we circle the conserved patterns given by blocks of identical residues:

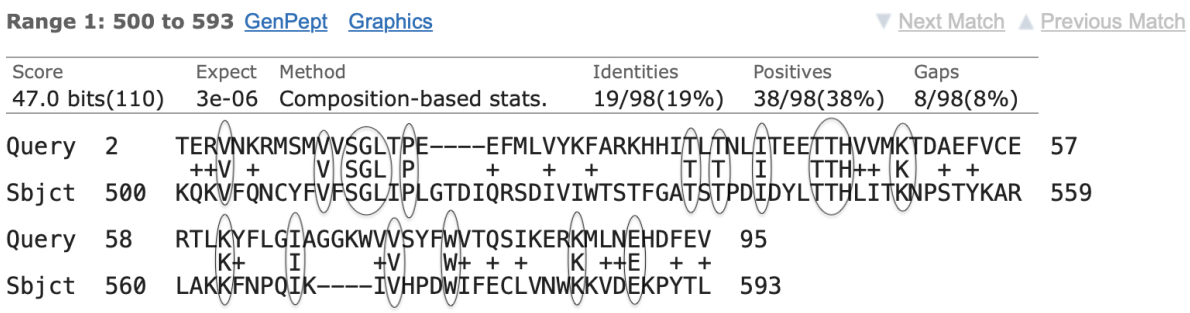

Figure 1

To find the identical negatively charged residues, we must search for acid amino acid, and there are only Aspartate (D) and Glutammate (E) that are negatively charged. In this alignment we count for only one identical negatively charged residues, that is the last identical residue (Glutammate).

For the similar conserved positively charged residues, we have to search for possible combination of basic amino acid: Lysine (K), Arginine (R) and Histidine (H). We can find exactly four identical positively charged residues (three times identical Lysine and only one time identical Histidine). Looking instead for similar positively charged residues, we can only find Arginine with Lysine (in the image is written as R+K).

2 HMM

Given the following training sequence, to build an HMM we first of all define which column/s make which state, that means:

- Match($\square$)$\longrightarrow$ all character
- Insertion ($\diamond$)$\longrightarrow$ n°gaps $>$ n°character
- Deletion ($\circ$)$\longrightarrow$ n°gaps $<$ n°character. In this case we also consider a match state.

So the columns of the sequences:

```
-  C  A  -  -  -  T  A
T  C  A  A  C  T  G  A
-  C  A  C  -  -  G  A
G  G  A  -  -  -  C  A
-  C  C  G  -  -  G  A
```

can be identified in the following states:

- I_1
- M_1
- M_2
- M_3 and D_1
- I_2
- I_2
- M_4
- M_5

We then can draw the scheme of the HMM model. We start by a dummy state B that stands for "Begin" and since the first position we have an Insertion we immediately split the transition probability in the case in which we enter in the Insertion state (basically the first column have a nucleotide $\frac{2}{5}$ this happens) or we jump to the first match case (there is a gap in the first column: $1 - \frac{2}{5}$ or simply $\frac{3}{5}$). Then inside the insertion state we have one time Thymine and one time Guanine so the probability is 50% for each of them. In the M_1 state instead we Cytosine four time so the probability is $\frac{4}{5} = 0.8$.

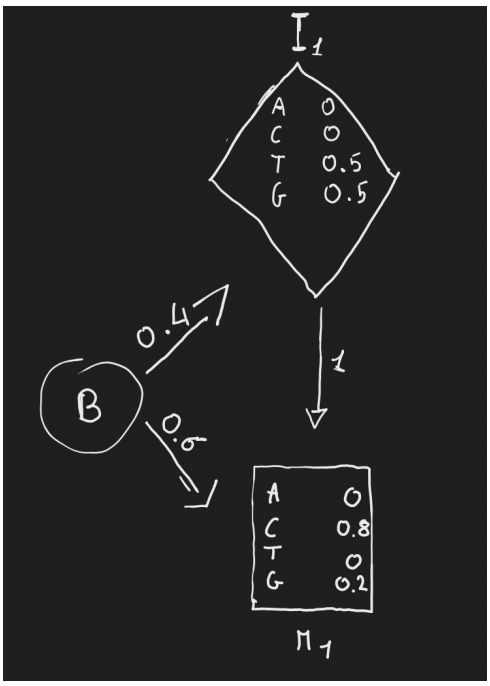

Figure 2

We then continue in the state M_2 with probability one (no other options) and after that we encounter a Deletion state D_1. In this case the Nucleotide that are present in this column enter in the state M_3 created with the state D_1 with probability in this case of $\frac{3}{5} = 0.6$. Inside the M_3 however, we do not evaluate the emission probability with 5 at the denominator, but with 3. In fact inside the M_3 there are only the nucleotide that are present in the column, since the other goes in the D_1 state. So we have 33% for Adenine, Guanine, and Cytosine.

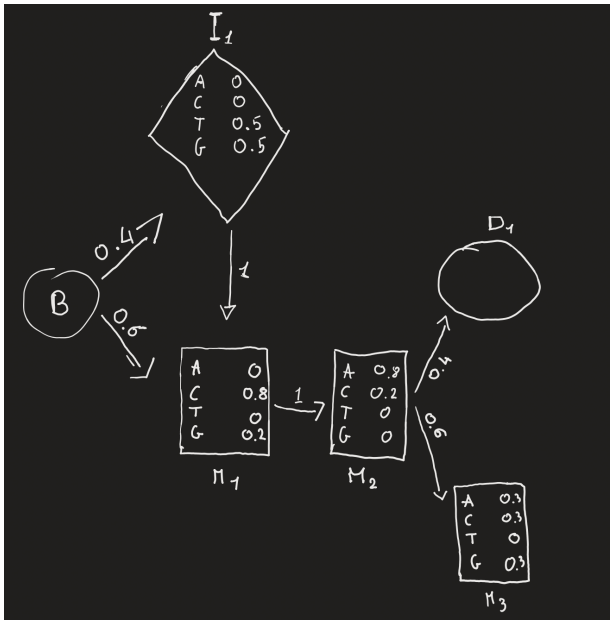

Figure 3

From the D_1 state we move with probability one to M_4 state (there are only gap before that) but the element in the state M_3 could either jump directly to the M_4 state (as 2 of them do, so the transition probability is $\frac{2}{3} = 0.66$) or enter in the Insertion state I_2 ($\frac{1}{3} = 0.33$). Inside the Insertion state this time, since we have more than one gap, we need to evaluate the e, that is the transition probability that the Insertion state enter again inside itself and since we have only two jumps inside the Insertion state, one must be $I_2 \longrightarrow I_2$ and the other $I_2 \longrightarrow M_4$ so that the value of $e = \frac{1}{2} = 0.5$ and same value have the transition probability from I_2 to M_4

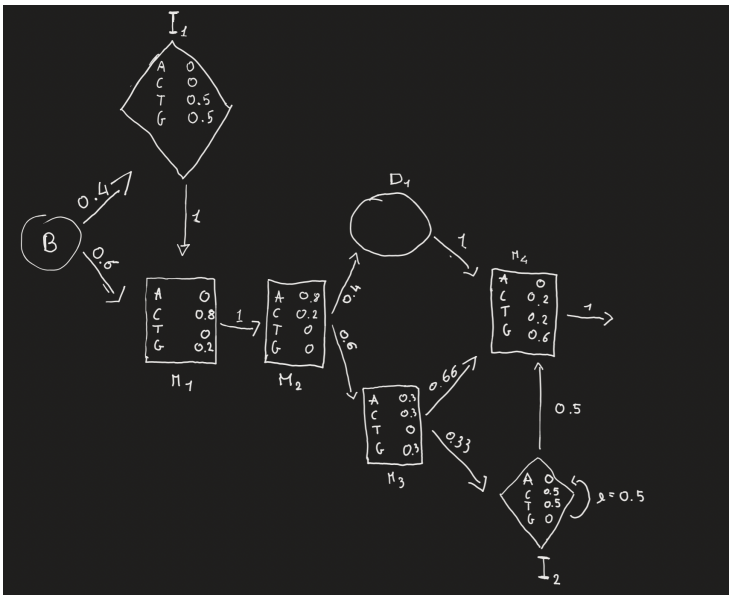

Figure 4

The last passages are straightforward, since the transition probability is 1 from M_4 to M_5 and the emission probability is easy to evaluate. We also need to finish after the last state with a dummy state E that stands for "End".

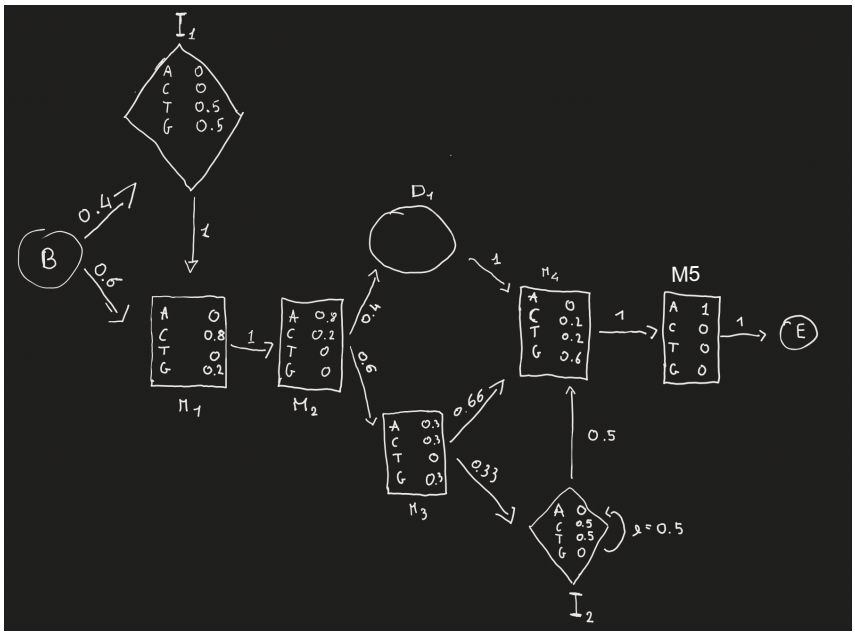

Figure 5